

Advanced applied econometrics

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Next week

- 1 next and final lecture: 10.7; 9-12
- 2 prepare Blundell et al. (2016)

Plan for today

- 1 Keane & Wolpin: Set-up
 - Structure of Model
 - Identification
 - Model fit & fix
- 2 K&W: Solution and Estimation
 - compare to Rust
 - Simulate & Interpolating value function
- 3 Practical session: Homework (Maxi)

Motivation of Keane & Wolpin

What is the research question?

Main aims of paper

Explain **schooling & career choices** (“human K investment”)

- school
- white / blue collar work
- military
- home production

Policy Evaluations: Impact of tuition subsidy

Motivation of Keane & Wolpin

Assumptions ? Alternatives? Atheoretic answers?

What **choices** does paper model?

Rewards of choice

- How are the rewards of choices modelled?

Rewards of choice

(1) Wage

- occupation-specific skill heterogeneity $e_m(a)$
- schooling $g(a)$
- experience in occupation $x_m(a)$ quadratic
- unobserved endowments at age 16 $e_m(16)$
- **cross-experience** terms (what?)- see Section (III)

(2) non-monetary “rewards”

- **effort** cost of schooling
- **home** “production” (what?)

Unobserved heterogeneity

How modelled in Keane & Wolpin?

Unobserved heterogeneity (types)

Individual “fixed” effects very **interactive**

- e_{mk} are K types of initial endowments in diff occupations
- initial diffs create diff **comp advantage**
- link to initial schooling (initial conditions) ?

Schooling choices & technology

Schooling in model?

Schooling choices & technology

Costs of schooling

- monetary tuition fees (college & graduate)
- current period effort cost
- individual cost of schooling via $e_4(16)$

What are benefits of schooling?

State-space

What is state-space ?

State-space

At any point in time t , observe

- **past** choices
- initial **conditions**
- current **shocks**

... **how do individuals choose** what to do?

Choices

Choices made subject to

- draw shocks
- compare expected value of actions - Value functions.

(more on **solution method** in a minute)

Structure of Model

Assumptions, **Alternatives**, Atheoretic answers ?
What alternative assumptions could you imagine?

- Preferences
- Technology/ Market
- General structure - are processes missing?

Alternatives

Preferences

- non-monetary prefs over activities (edu ?) may change over time
- people care not only about themselves
 - people conform to **norms**
- people may care about risk
- people may be **myopic or backwardlooking**
- people may dislike specific occupation

Alternatives II

Technology

- information maybe imperfect
- hours choice may be possible

Market

- labour demand ?
- transitions not always **voluntary** (labor market frictions)
- single agent model (cf. BLP or search model)

Structure of Model

Assumptions, Alternatives, **Atheoretic answers** ?

What atheoretic strategies for this research question?

Less structural approaches

- 1 **OLS**
- 2 **Exogenous schooling & occupations**
Quasi-experimental IV- and panel literature.
- 3 **Static discrete choice**: model multinomial choice

Less structural approaches II

Estimates of rates of return to education (RORE) are...

Less structural approaches II

OLS-estimated returns to educ'n/experience/occupations:
...biased if **unobserved factors** influence...

- 1 schooling choice & earnings
- 2 **selection** into work & experience.
- 3 **selection** into occupations

Here: **model selection** into schooling, experience, occupational choice.

Alternative?

Identification

How is the model identified ?

Identification

Transitions between activities over age

- schooling becomes less frequent
- working increases.

Occupational differences in transition trends

- blue-collar work occurs often at low ages
- blue-collars hardly increase after age of 22

Earnings evolution over age & transitions

- voluntary transitions / persistence must be incentivised
- what about **non-monetary costs/rewards** ?

Identification

- Data: Cohort study - NLSY 1979 - 1988
- What can explain **wage growth** in occupation?
- What role for **measurement error** in wages ?
- What can explain **high persistence** in activities ?
- What can identify **discount factor** ?

Identification

What can explain **wage growth** in occupation?

- experience & age
- no job-to-job transitions (no wage variance in occup'n)
- no bargaining, no promotions

What role for **measurement error** in wages ?

- variance of measurement error **smooths predictions** (deal with outlier) - see FN 25.

Persistence?, Discount factor?

Percentage in state by age

How to get a **good fit**? see Figures 1-5 in paper.

Extended model: More persistence

Persistence in choices **stronger than model** predicts.

How to fix the fit ?

Extended model: More persistence

- 1 Occupation **switching costs**.
Drop in earnings on leaving sector for a year
- 2 Occupation-specific **non-monetary costs** - interpretation ?
- 3 age-specific home term (young people couch premium ?)
or involuntary unemployment ?
- 4 “psychic” **graduation effect**
non-monetary schooling return

Results of augmented model

- 1 Importance of **initial endowments (types)** for w-var
- 2 Measurement error “accounts for” 40% of total w-var
- 3 Experience terms are important and heterogenous.
- 4 **No diploma effect** on wages (warm glow ? employment ?)
- 5 Monetary **job search costs** are significant.
- 6 How to interpret Table 12?

Solution methods

Compare **Rust** to **Keane & Wolpin** solution methods.

Keane & Wolpin

Dynamic Bellman equation:

$$\begin{aligned} V_t(S_{i,t}, \theta) &= \max_{d(S_{i,t})} E \left[\sum_{\tau=t}^T \beta^{t-\tau} u_{\tau}(d_{\tau}, X_{i,\tau}, \theta) + \varepsilon \right] \\ &= \max_{d(S_{i,t})} [u_t(d_t, X_{i,t}, \theta) + \beta E [V_{t+1}(S_{i,t}, \theta) | X_{i,t}, d_t]] \end{aligned}$$

Conditional independence:

- Dynamic term $\beta E [V_{t+1}(S_{i,t}, \theta) | X_{i,t}, d_t]$ indep of shock
- choice-reward combinations dep on **current state-space**

How to calculate $\beta E [V_{t+1}(S_{i,t}, \theta) | X_{i,t}, d_t]$?

Solving for values

Infinite horizon (Rust)

- seek **fixed point**

Finite horizon (Keane & Wolpin)

- **backward induction**

Backward induction

(1) **Optimal policy** δ^* for **final period** problem simple

$$V_t(S_{i,t}, \theta) = \max_{d(S_{i,t})} [u_t(d_t, X_{i,t}, \theta)]$$
$$\delta_T^*(S_{i,T}, \theta) = \operatorname{argmax}_{d_T} [u_T(d_T, X_{i,T}, \theta) + \varepsilon_{i,d_T}]$$

(2) in period $T - 1$

$$E[V_t(S_{i,t}, \theta) | X_{i,T-1}, d_{i,T-1}] =$$
$$\int \int [u_T(\delta_T^*(S_{i,T}, \theta), X_{i,t}, \theta) + \varepsilon_{i,\delta_T^*}] dF_\varepsilon dF_X(X_T | X_{i,T-1}, d_{T-1})$$

What to do about those integrals?

Backward induction II

What to do about **those integrals**?

- 1 Integrate idiosyncratic shock: assume ε **is EV(1)**

$$\int \left[u_T(\cdot), X_{i,t}, \theta \right) + \varepsilon_{i,\delta_T^*} \right] dF_\varepsilon = \log \left(\sum_{d_T} e^{u_T(d_T, X_{i,T}, \theta)} \right) + \gamma$$

- 2 Integrate **future state space** (over $F(X_T | X_{i,T-1}, d_{T-1})$)?

Backward induction II

Theory: **Evaluate** choice prob's at **all points**.

- if $X_{i,t}$ has 5 binary variables
- if $X_{i,t}$ has 10 variables with 3 values

By how much does state space grow?

Backward induction III

Theory: **Evaluate** choice prob's at **all points**.

- if $X_{i,t}$ has 5 binary variables
= $2^5=32$ values for each t .
- if $X_{i,t}$ has 10 variables with 3 values
= $3^{10}=59,049$ values for each t .

Numerical challenge increases (fast!) in state space

Interpolating

Backward induction in practice:

- 1 **Solve at subset** of grid points
- 2 **Impute value** (interpolate) at other points

With optimal policy at $T - 1$, go to $T - 2$...

Keane & Wolpin interpolate **both integrals** - why?

- 1 Draw R vectors of shocks (one for each alternative)
 $\varepsilon_{1,T}, \dots, \varepsilon_{M,T}$ for T .
- 2 Evaluate the maximum for each draw of R .
- 3 Average of R draws: expected maximum (EMAX)

$$V_T(x) = \frac{1}{R} \sum_{r=1}^R \max_d u_T(d, x, \varepsilon_T^r(d))$$

- 4 then calculate $V_{T-1}(x)$ in $T - 1$

$$= \frac{1}{R} \sum_{r=1}^R \max_d \left[u_{T-1}(d, x, \varepsilon_{T-1}^r(d)) + \beta \sum_x i_{i,T} V_T(x_{i,T}) F(x_i, T | x_{i,T-1}, d) \right]$$

This simulates EMAX for R specific points ε .

Simulate & Interpolate II

Select R **grid-points** (specific values of X) and **impute EMAX for others**, e.g.

$$V_t(x_1 = 1, x_2 = 10, x_3 = \ln(1000)) = 125$$

$$V_t(x_1 = 2, x_2 = 100, x_3 = \ln(2000)) = 500$$

Establish link between **EMAX and state space**

$$V_t(x_{k,t}) = \alpha_1 x + \alpha_2 x^2 + \dots + \xi_{k,t}$$

Simulate & Interpolate (last slide)

Then, estimate predicted EMAX for actual individuals using $\hat{\alpha}$.

$$\hat{V}_t(x_{i,t}) = \hat{\alpha}_1 x_{i,t} + \hat{\alpha}_2 x^2 + \dots \quad (1)$$

10 minutes break